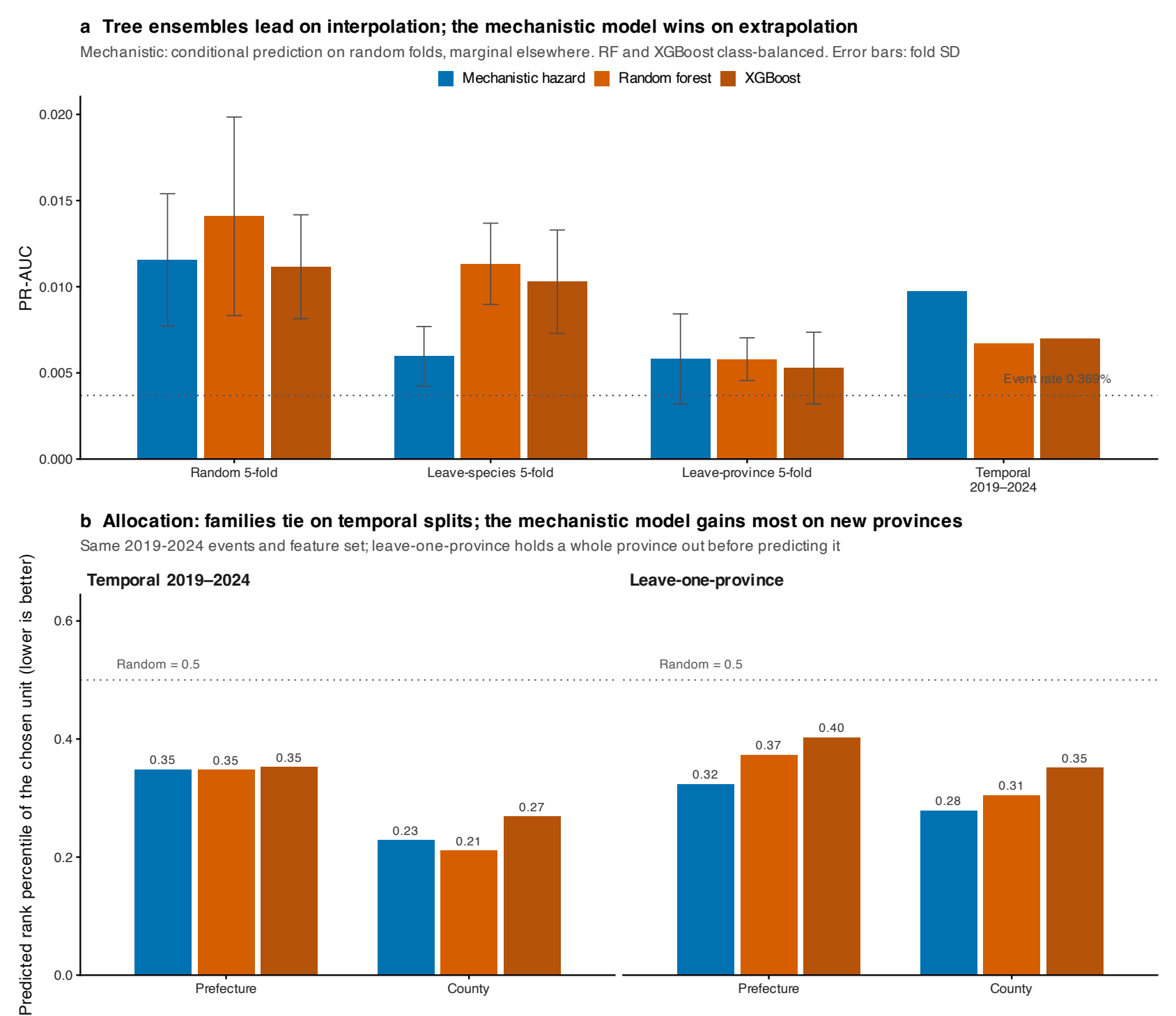
a Tree ensembles lead on interpolation; the mechanistic model wins on extrapolation
Mechanistic: conditional prediction on random folds, marginal elsewhere. RF and XGBoost class-balanced. Error bars: fold SD
Mechanistic hazard
Random forest
XGBoost
PR-AUC
0.020
0.015
0.010
0.005
0.000
Event rate 0.369%
Random 5-fold
Leave-species 5-fold
Leave-province 5-fold
Temporal 2019–2024
b Allocation: families tie on temporal splits; the mechanistic model gains most on new provinces
Same 2019-2024 events and feature set; leave-one-province holds a whole province out before predicting it
Temporal 2019–2024
Leave-one-province
Predicted rank percentile of the chosen unit (lower is better)
0.6
0.4
0.2
0.0
Random = 0.5
Random = 0.5
0.35
0.35
0.35
0.23
0.21
0.27
0.32
0.37
0.40
0.28
0.31
0.35
Prefecture
County
Prefecture
County